ARJUNA JEE 2027

Lecture -01 | Mathematics

PERMUTATIONS AND COMBINATIONS

By- Sachin Jakhar Sir



Topics *to be covered*

A **Graphical Transformation**

B **Fundamental Principle of Counting**

C **Question Practice**

D



H.W. Discussion

Acidic Question

If $\cos^3\theta + \cos^3(120^\circ + \theta) + \cos^3(240^\circ + \theta) = \frac{a}{b}\cos 3\theta$, where '$a$' and '$b$' are relative prime, then the value $a + b$ is

$$M = \cos\theta + \cos(\overbrace{120^\circ}^{A} + \overbrace{\theta}^{B}) + \cos(240^\circ + \theta) = 0.$$

$$M = \cancel{\cos\theta} \cancel{-\frac{1}{2}\cos\theta} \cancel{-\frac{\sqrt{3}}{2}\sin\theta} \cancel{+\frac{-1}{2}\cos\theta} \cancel{+\frac{\sqrt{3}}{2}\sin\theta}$$

$$= 0.$$

$\cos 120^\circ = -\frac{1}{2}$

$\sin 120^\circ = +\frac{\sqrt{3}}{2}$

$\cos 240^\circ = -\frac{1}{2}$

$\sin 240^\circ = -\frac{\sqrt{3}}{2}$

Acidic Question

If $\cos^3\theta + \cos^3(120^\circ + \theta) + \cos^3(240^\circ + \theta) = \frac{a}{b}\cos 3\theta$, where '$a$' and '$b$' are relative prime, then the value $a + b$ is

$180^\circ + \theta - 60^\circ$ $\quad$ $180^\circ + 60^\circ + \theta$

7 Ans.

$x^3 + y^3 + z^3 = 3xyz$ $\quad$ $x + y + z = 0$

$= 3\cos\theta \cdot (\cos(\theta - 60^\circ))(\cos(60^\circ + \theta))$ $\quad$ ($60^\circ - \theta$)

$= \frac{3}{4}\cos 3\theta$

Question

If $f(\theta) = \sin^2\theta + \sin^2\left(\theta + \frac{2\pi}{3}\right) + \sin^2\left(\theta + \frac{4\pi}{3}\right)$, then $f\left(\frac{\pi}{15}\right)$ is equal to

A $\frac{2}{3}$

B $\frac{3}{2}$

C $\frac{1}{3}$

D $\frac{1}{2}$

$$\frac{1-\cos 2\theta}{2} + \frac{1-\cos\left(2\theta + \frac{4\pi}{3}\right)}{2} + \frac{1-\cos\left(2\theta + \frac{8\pi}{3}\right)}{2}$$

Acidic Question

The value of the expression $\frac{\sin 7\alpha+6\sin 5\alpha+17\sin 3\alpha+12\sin\alpha}{\sin 6\alpha+5\sin 4\alpha+12\sin 2\alpha}$, where $\alpha=\frac{\pi}{5}$ is equal to

A $\frac{\sqrt{5}-1}{4}$

B $\frac{\sqrt{5}+1}{4}$

C $\frac{\sqrt{5}+1}{2}$

D $\frac{\sqrt{5}-1}{2}$

$2\cos 36^\circ = 2\left(\frac{\sqrt{5}+1}{4}\right)$

$N^r = \sin 7\alpha + \sin 5\alpha + 5\sin 5\alpha + 5\sin 3\alpha + 12\sin 3\alpha + 12\sin\alpha$

$2\sin 6\alpha\cos\alpha + 5(2\sin 4\alpha\cdot\cos\alpha) + 12(2\sin 2\alpha\cos\alpha)$

$N^r = 2\cos\alpha\,(\sin 6\alpha + 5\sin 4\alpha + 12\sin 2\alpha)$ → D^r

$\frac{N^r}{D^r} = 2\cos\alpha$

Question [JEE (Adv.)-2011]

The positive integer value of $n > 3$ satisfying the equation

$$\frac{1}{\sin\left(\frac{\pi}{n}\right)} = \frac{1}{\sin\left(\frac{2\pi}{n}\right)} + \frac{1}{\sin\left(\frac{3\pi}{n}\right)}$$ is





GRAPHICAL TRANSFORMATION

If graph of $y = f(x)$ is known then :

We can draw graph of ⇒ $|f(x)|$, $f(|x|)$, $-f(x)$, $f(-x)$

$f(x) \pm k$, $f(x \pm k)$, $kf(x)$, $f(kx)$

$|y| = f(x)$

RAPID FIRE ROUND

Draw the graphs of:

(i) $y = |f(x)|$

(ii) $y = f(|x|)$

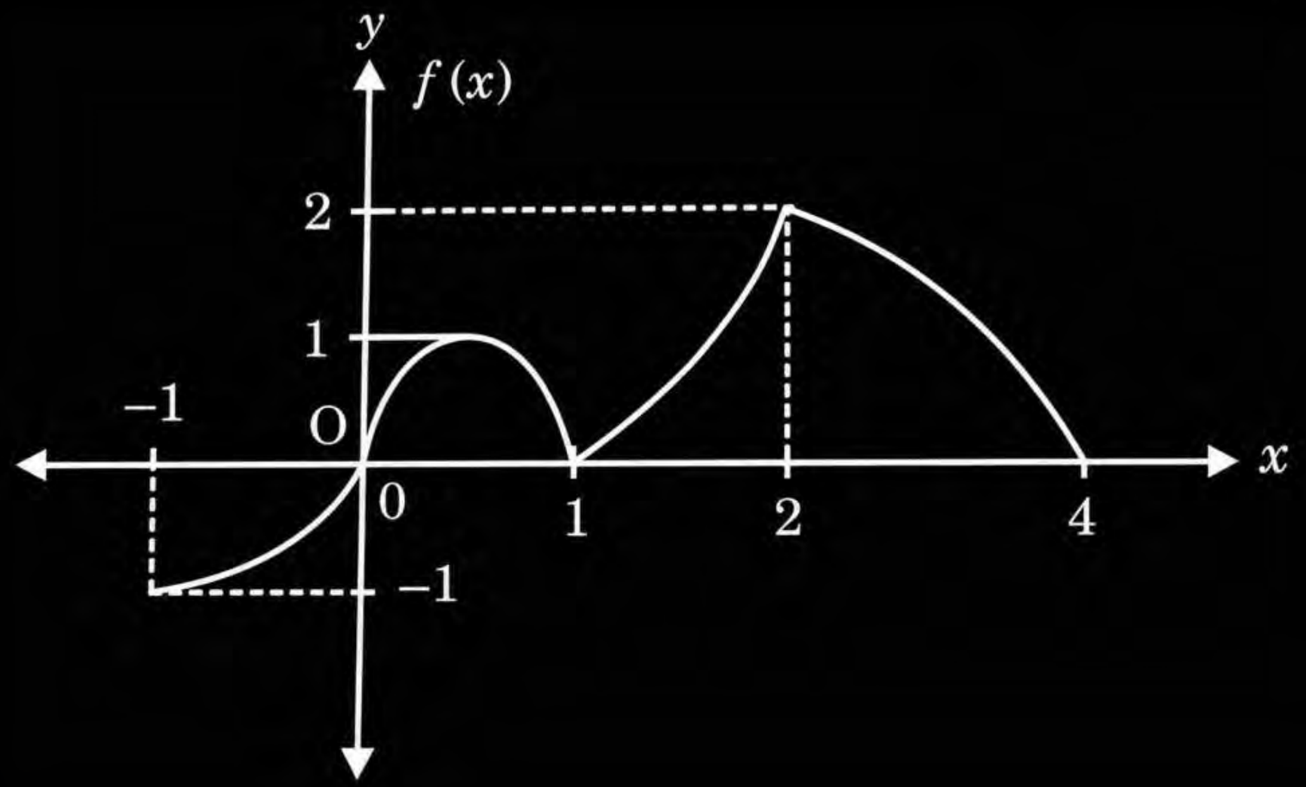
(iii) $y = |f(|x|)|$

"Same".

(iv) $y = -f(x)$

(v) $y = f(-|x|)$





Note :

$y = f(x)$

$\downarrow$

$y = f(-x)$

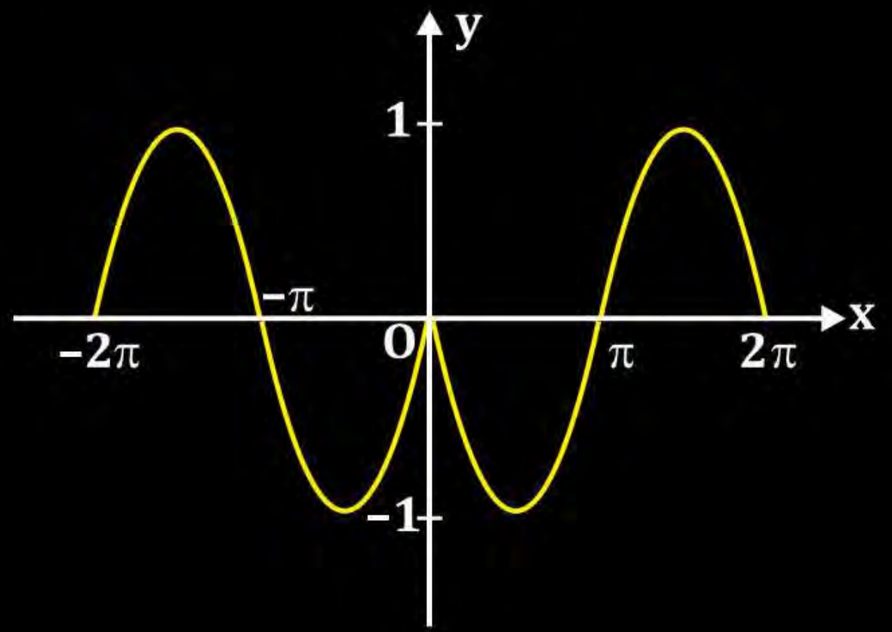
$\downarrow$

$y = f(-|x|)$

Question

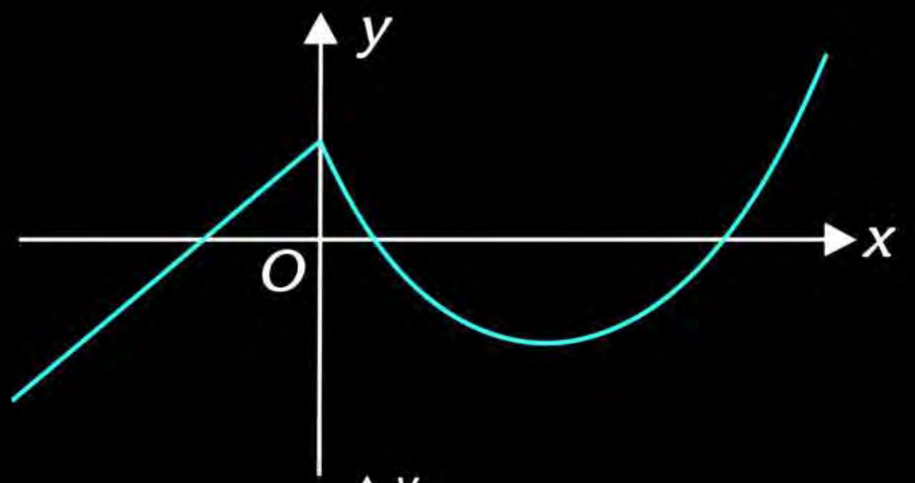
Which function corresponds to the given graph?

(A) $y = |\sin x|$ **(B)** $y = \sin|x|$ **(C)** $y = -|\sin x|$ **(D)** $y = -\sin|x|$



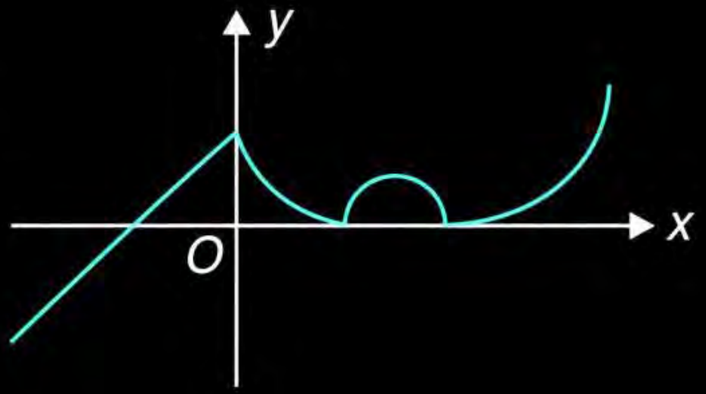
Question

The graph of $y = f(x)$ is shown in figure :

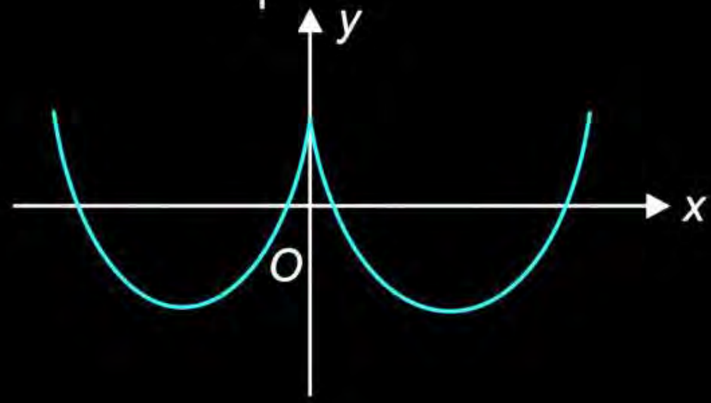


Which of the following best represent the graph of $y = f(-|x|)$?

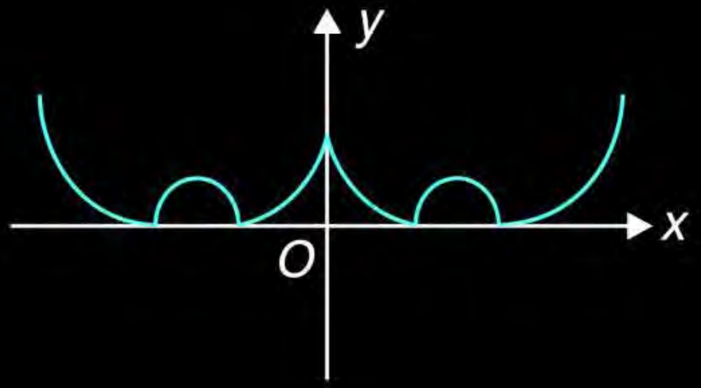
(A)



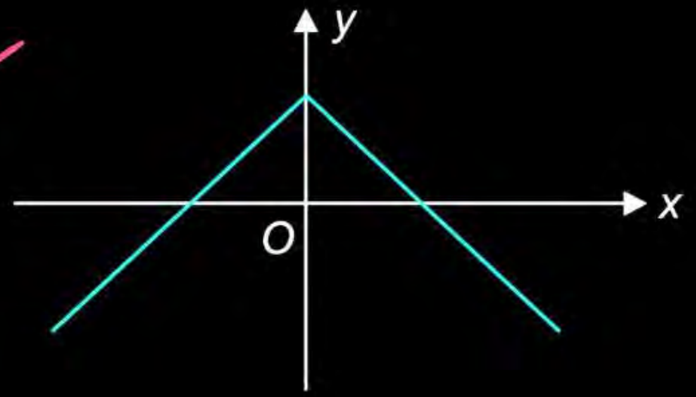
(B)



(C)



(D)





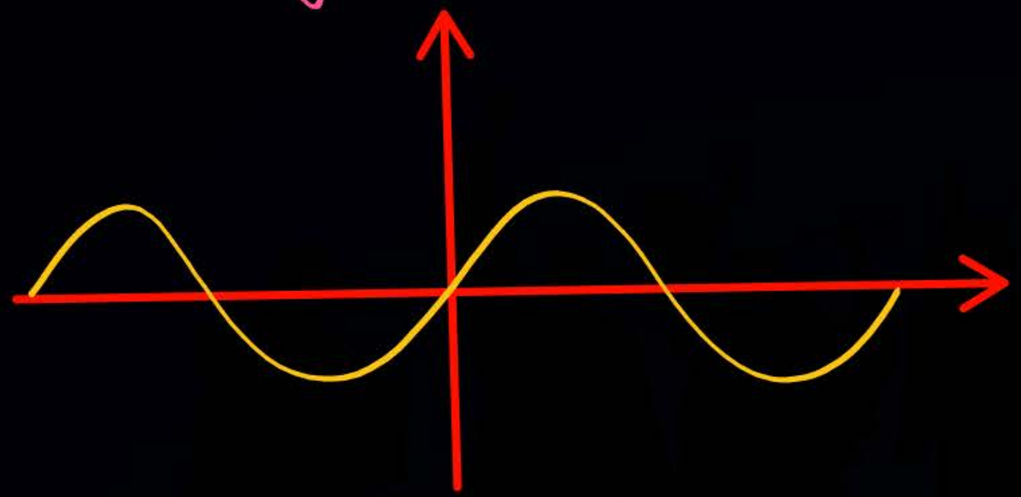
GRAPHICAL TRANSFORMATION

T-VI : To Draw $y = f(x \pm k)$

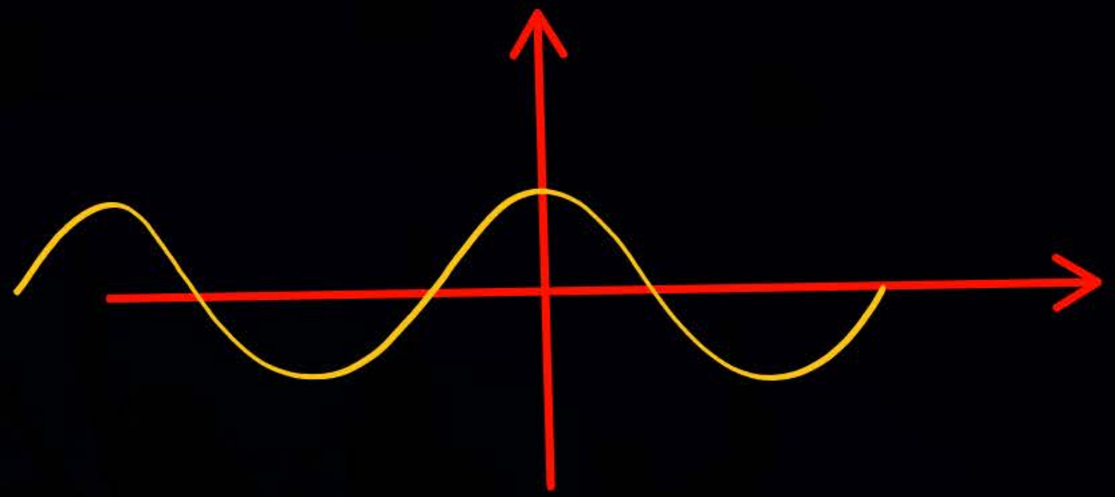
For $f(x + k)$: Shift whole graph by 'k' units in left direction (along X-axis)

For $f(x - k)$: Shift whole graph by 'k' units in right direction (along X-axis)

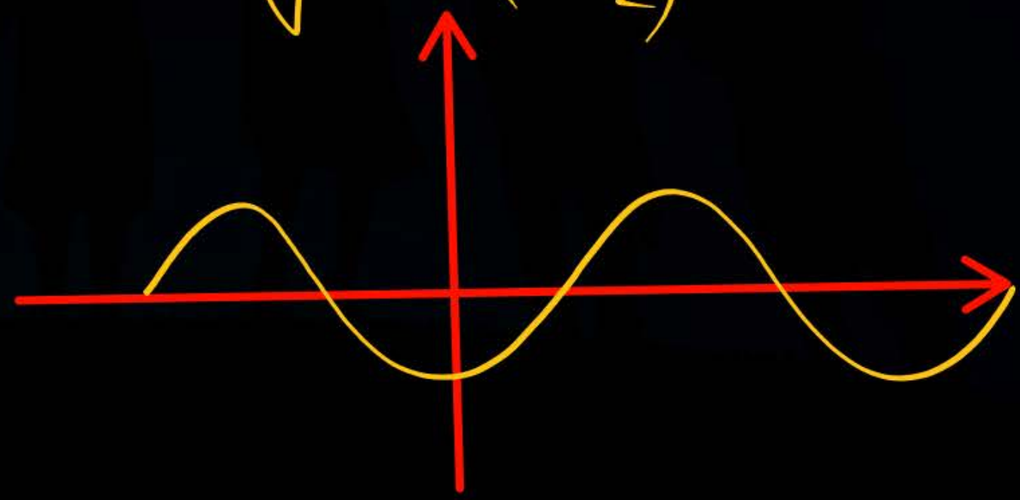
$y = \sin(x)$

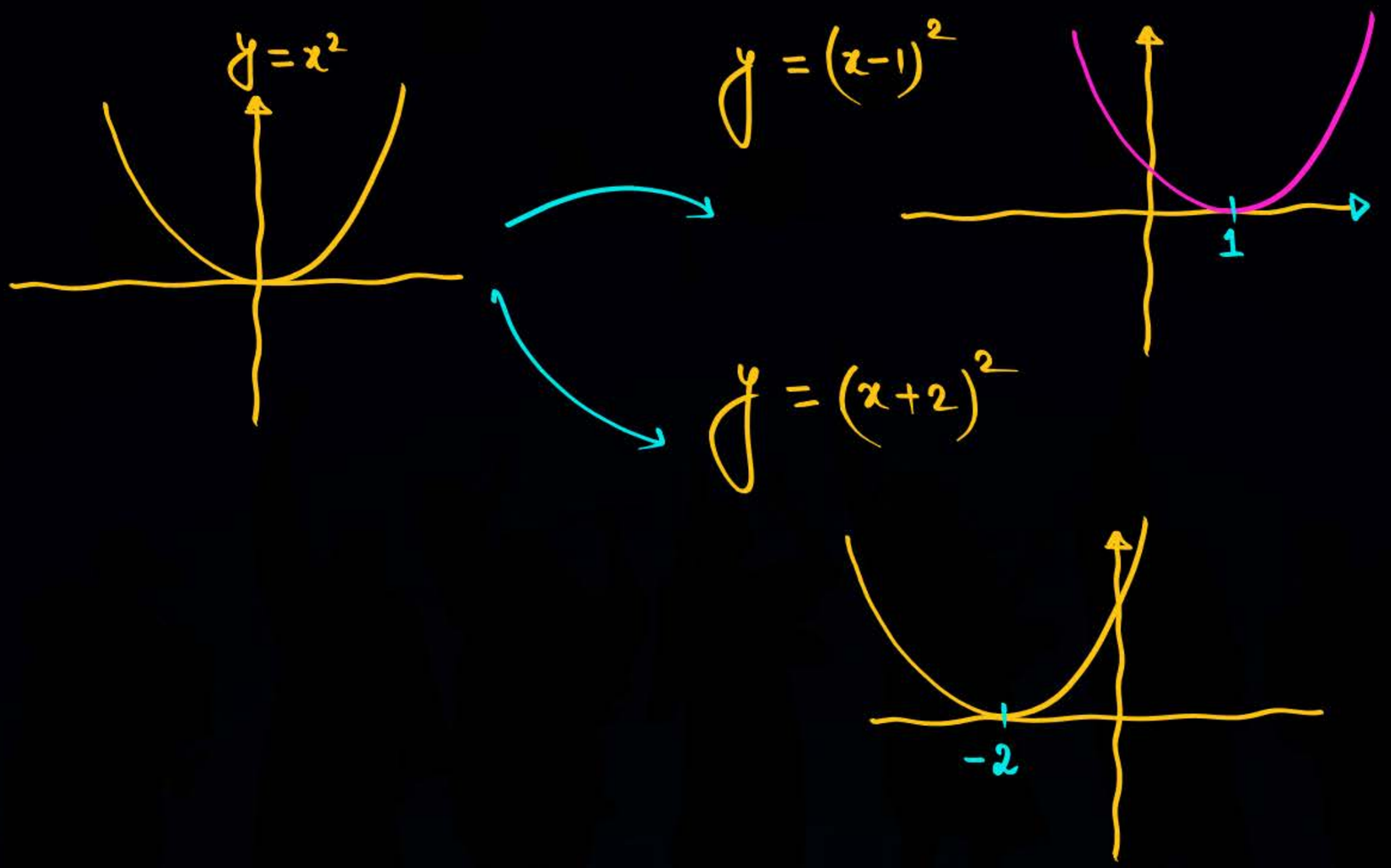


$y = \sin\left(x + \frac{\pi}{2}\right) = \cos x$



$y = \sin\left(x - \frac{\pi}{2}\right) = -\sin\left(\frac{\pi}{2} - x\right) = -\cos x$




$y = x^2$
$y = (x-1)^2$
1
$y = (x+2)^2$
-2



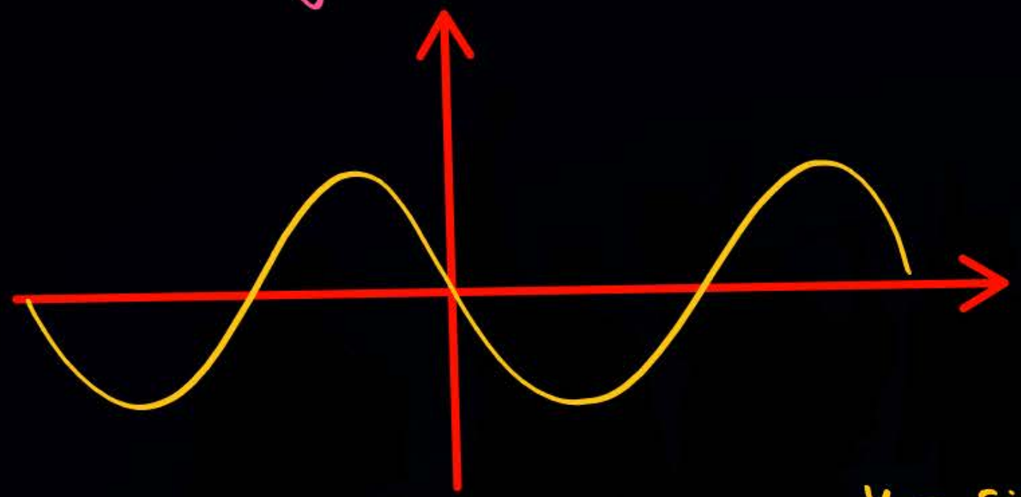
GRAPHICAL TRANSFORMATION

To Draw $y = f(-x \pm k)$

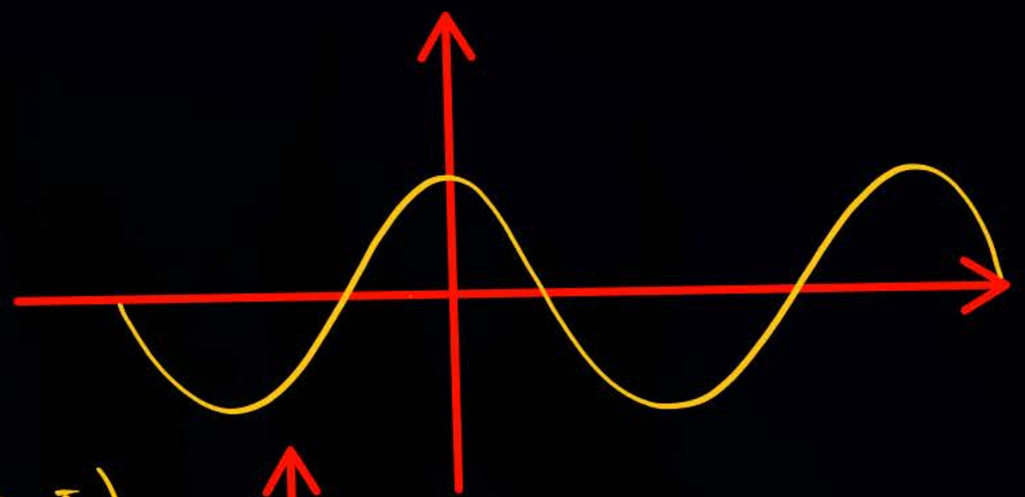
For $f(-x + k)$: Shift whole graph by 'k' units in right direction (along X-axis)

For $f(-x - k)$: Shift whole graph by 'k' units in left direction (along X-axis)

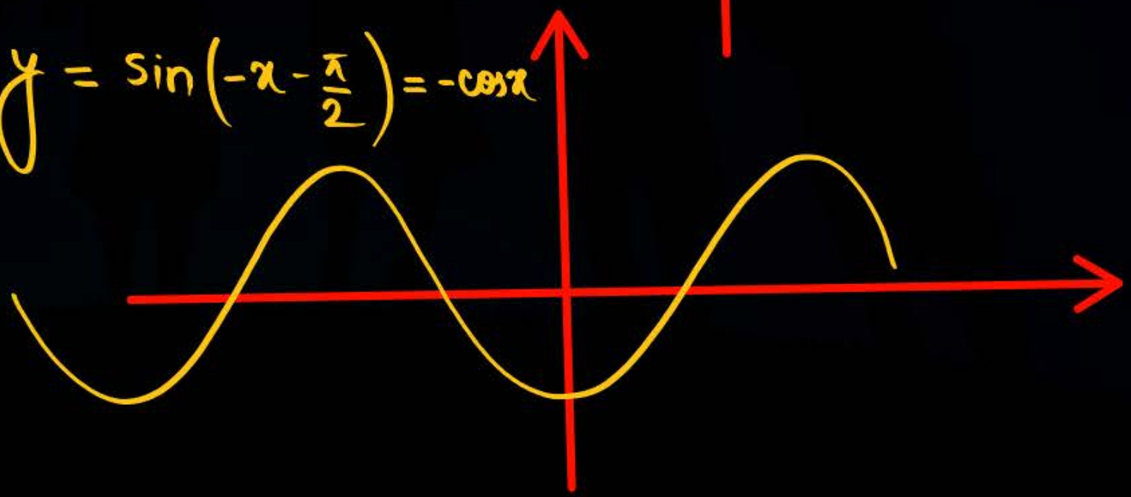
$y = \sin(-x)$



$y = \sin\left(-x + \frac{\pi}{2}\right) = \cos x$



$y = \sin\left(-x - \frac{\pi}{2}\right) = -\cos x$





GRAPHICAL TRANSFORMATION

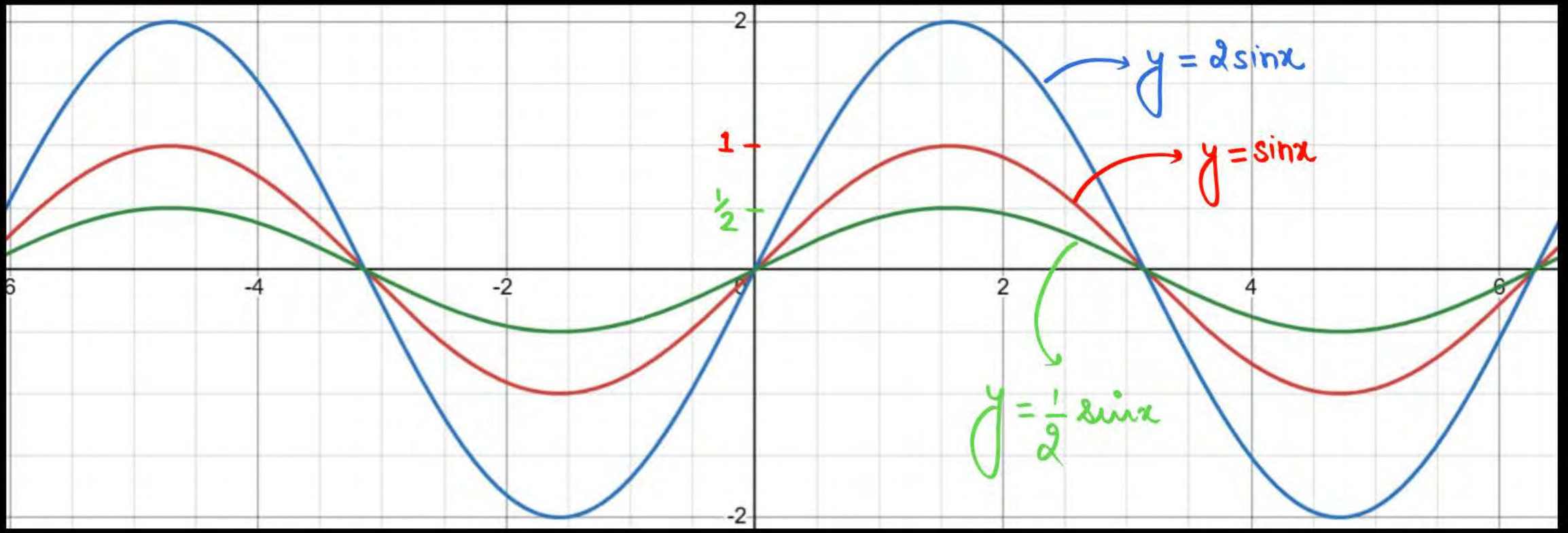
T-VII : To Draw y = kf(x), $(k > 0)$

Multiply every number on Y-axis by 'k'.

As a result : If $k > 1$, graph will stretch along Y-axis.

If $0 < k < 1$, graph will compress along Y-axis.

Example





GRAPHICAL TRANSFORMATION

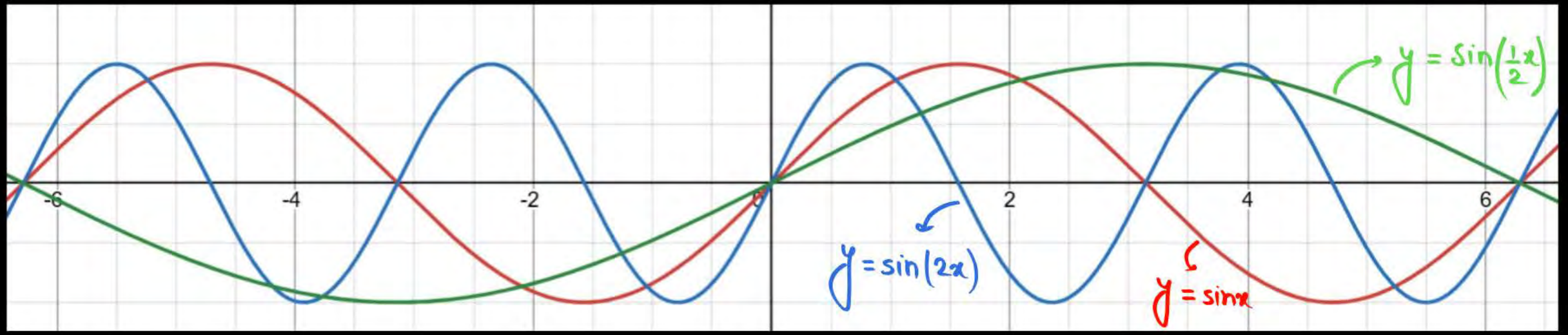
T-VIII : To Draw y = f(kx), $(k > 0)$

Divide every number on X-axis by 'k'.

As a result : If $k > 1$, graph will compress along X-axis.

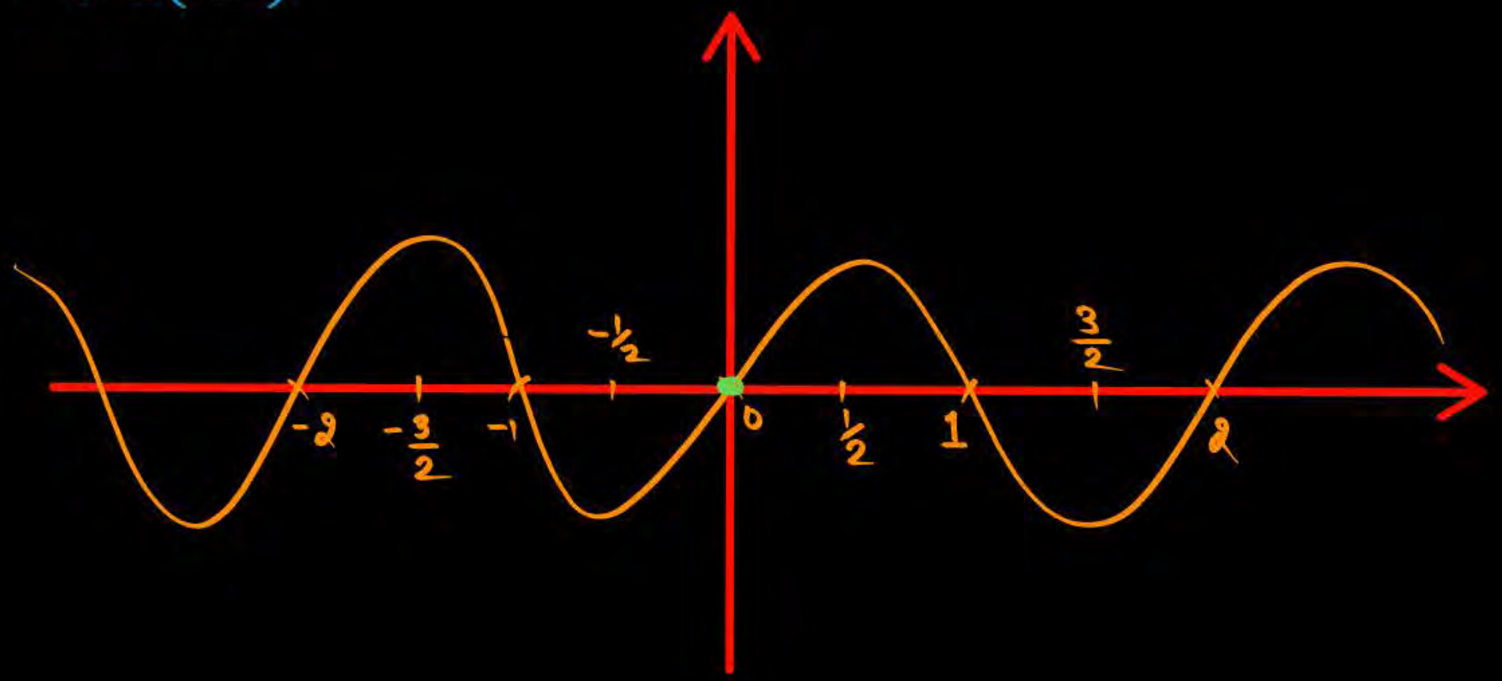
If $0 < k < 1$, graph will stretch along X-axis.

Example



Example

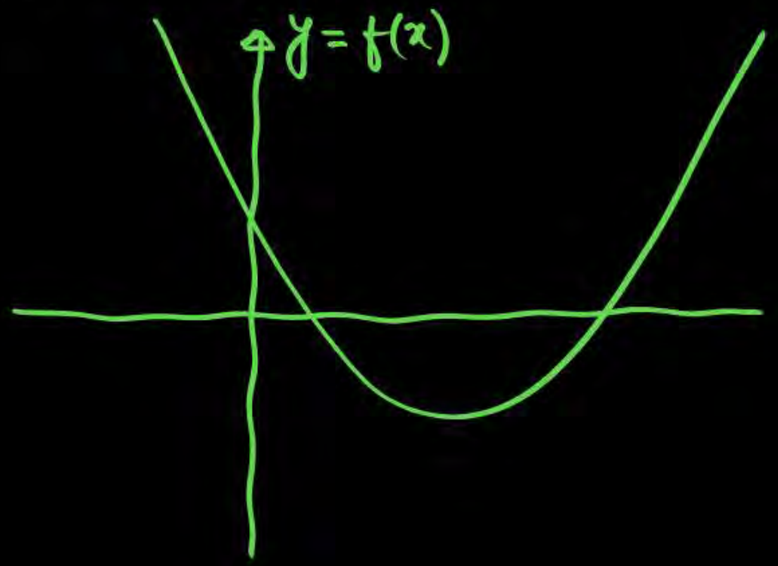
Draw : $y = \sin(\pi x)$

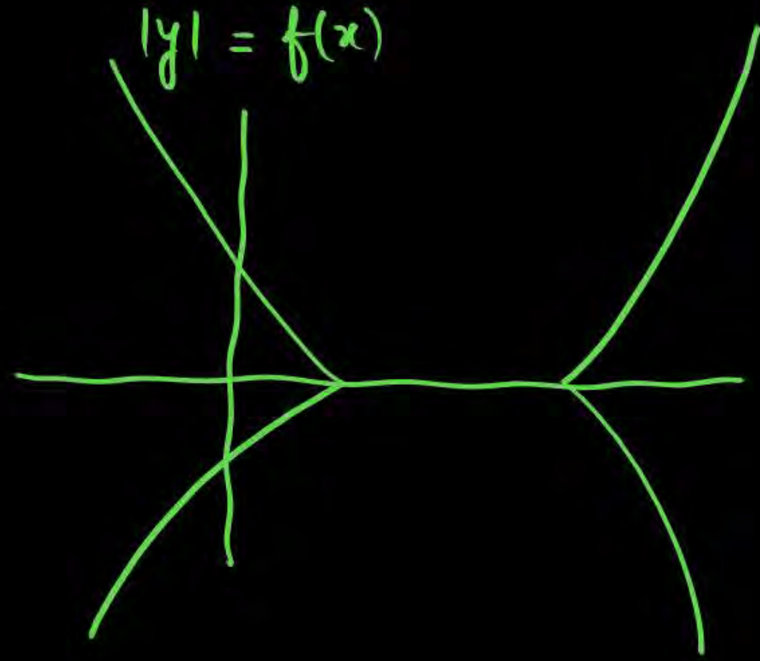


GRAPHICAL TRANSFORMATION

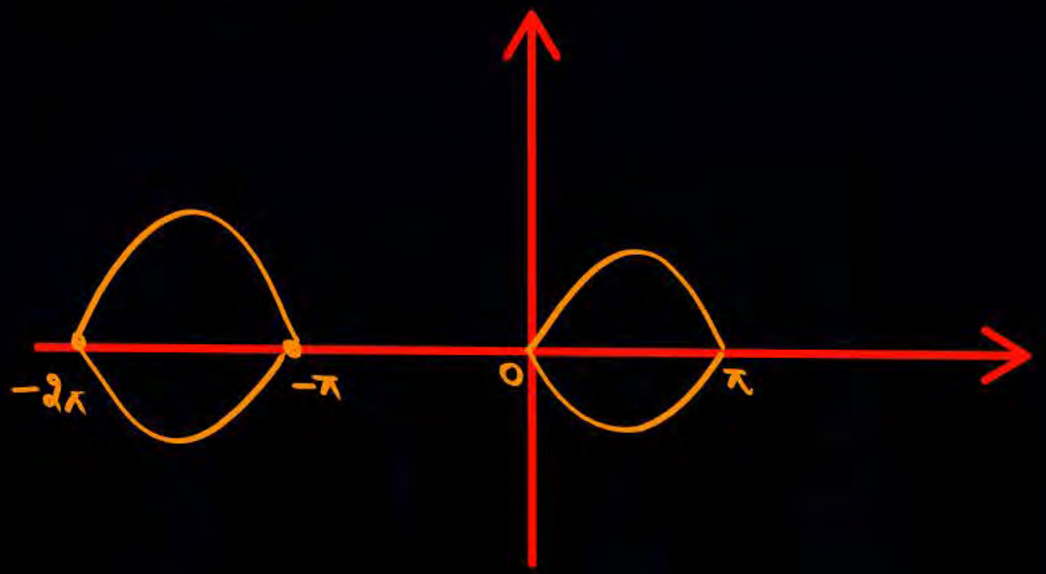
T-IX : To Draw |y| = f(x)

Remove part of the graph below X-axis, now take image of portion above X-axis in X-axis & keep both.





$|y| = \sin x$.



Example

Draw following graphs using multiple transformation:

(i) $y = |\ln|x||$

(ii) $y = \big||x| - 2| - 3\big|$

(iii) $y = e^{-|x|} + 2$

(iv) $y = -\{-x\}$

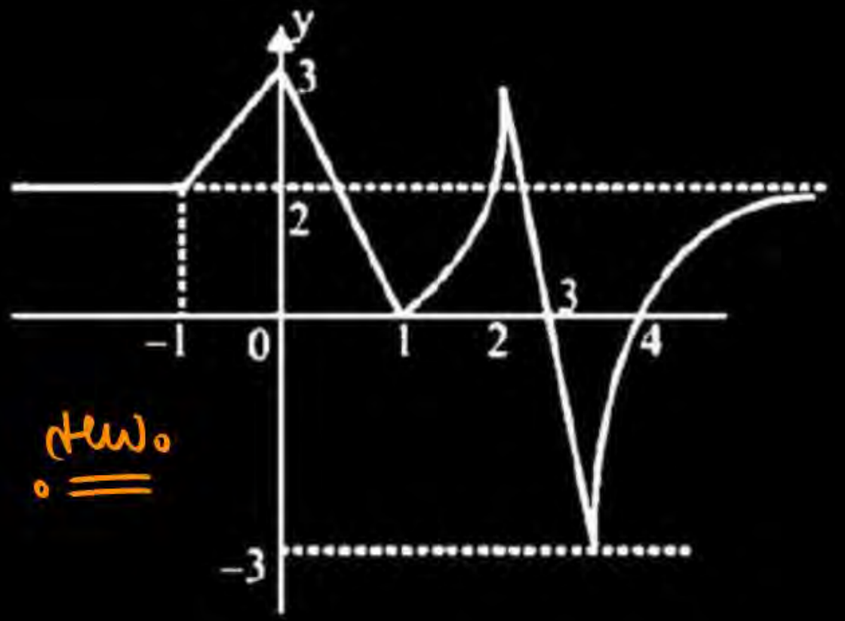
(v) $|y| = 3 + \sqrt{2 - x}$

(vi) $y = |x^2 - 5|x| + 6|$

new

Question

If graph of a function $f(x)$ is shown as following then number of solutions of equation $|f(|x|)| = 2$ is equal to



A 3

B 6

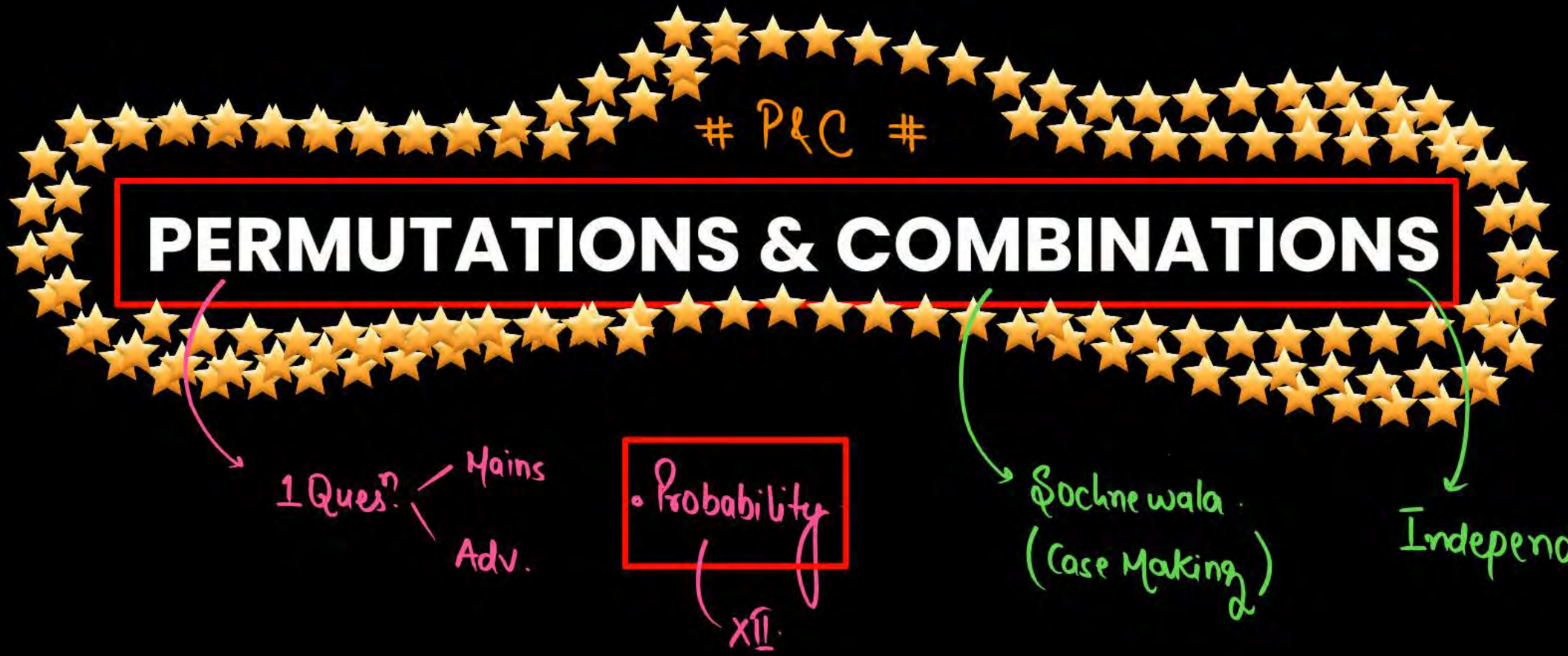
C 10

D Infinite

Question

Number of solutions of equation $\left|x^2 - |x|\right| = \ln|x|$ is equal to

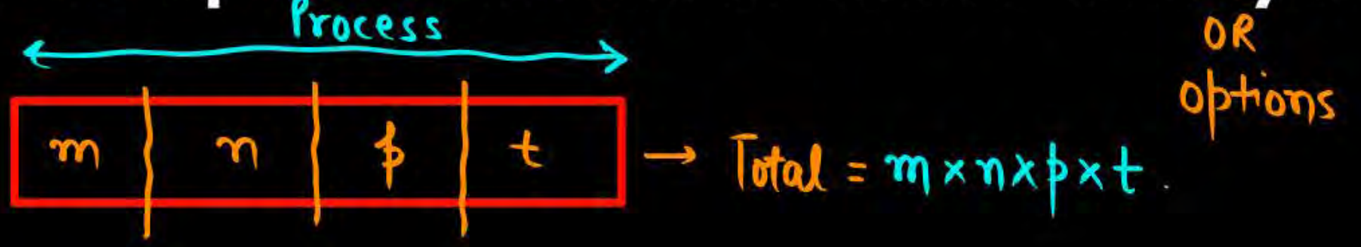



P&C
PERMUTATIONS & COMBINATIONS
1 Ques^n
Mains
Adv.
• Probability
XII
Sochne wala
(Case Making)
Independent

FUNDAMENTAL PRINCIPLE OF COUNTING

Multiplication Principle: (AND)

Jab tak process complete na ho tab tk number of ways ko multiply krte rho.



Addition Principle: (OR)

Agr given process 'm' tariko se complete ho gya hai & ussi process ko complete krne ke 'n' aur tarike hain, to total ways = (m + n) ways



Question

Let a person "Little Golu" has 3 different shirts, 2 different Jeans & 4 different belts. Then find number of ways of getting ready.

Shirt → S_1 S_2 S_3

Jeans → J_1 J_2

Belt → B_1 B_2 B_3 B_4.



Shirt → 3
Jeans → 2
Belt → 4

fnf = 3×2×4
= 24 ways

Question

Let a person "Little Golu" has 3 different shirts, 2 different Jeans & 4 different belts. Then find number of ways of getting ready.

	Shirt	Jean	Belt
	S_1	J_1	B_1
	S_1	J_1	B_2
	S_1	J_1	B_3
	S_1	J_1	B_4
	S_1	J_2	B_1
	S_1	J_2	B_2
	S_1	J_2	B_3
	S_1	J_2	B_4
	S_2	⋮	⋮
	⋮		



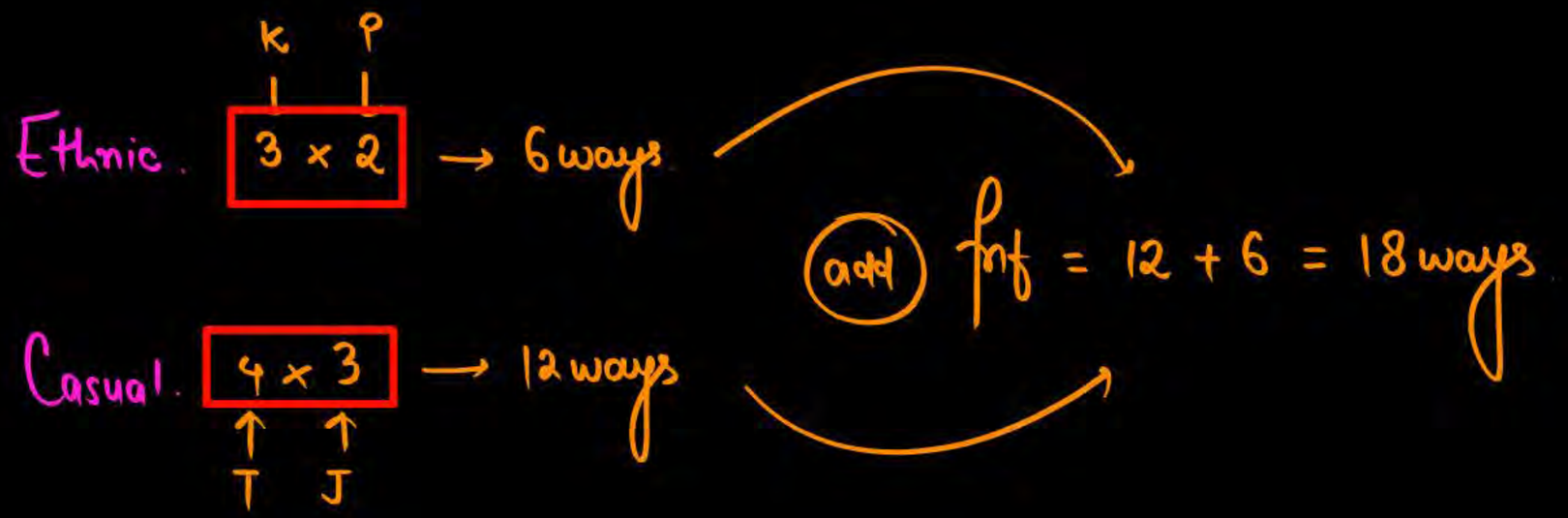
Shirt → 3
Jeans → 2
Belt → 4

$fnf = 3 \times 2 \times 4$
$= 24$ ways.

Question

Consider 3 different Kurtas, 2 different Pajamas, 4 different T-Shirts & 3 different Jeans.

In how many ways can little Golu dress up for an event either fully ethnic or fully casual.



Question

There are 7 flags of different colour. Find the number of different signals that can be transmitted by the use of 2 flags one above the other.



Total = $7 \times 6 = 42$.

G G G G G G
B Bl. W Y O P

B B B B B B
G Bl. W Y. O P

Question

Section-B of JEE-Adv. contains six multiple choice questions in which one or more than one options can be correct. Find total number of ways to attempt these 6 questions:

(i) Each question should be answered.

(ii) One can leave the question unattempted.

$(16)^6$

Q1 Q2 Q3 Q4 Q5 Q6

15 15 15 15 15 15

Total $= (15)^6$

1 Ques.

Multiple Correct

Q1. ○A ○B ○C ○D

2 opt. × 2 opt. × 2 opt. × 2 opt. = (16) ways

- not-attempted.
- A
- B
- C
- D
- AB, AC, AD.
- BC, BD, CD.
- ABC, ACD, ABD, BCD.
- ABCD.

Sec-B

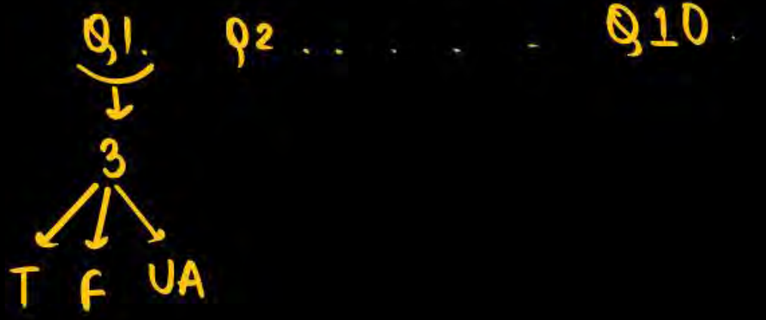
Q1	Q2	Q3	
16	16	16	$\rightarrow (16)^3$
A	A	$\overline{A}$ B C D AB	16 ways.
A	B	$\overline{A}$ B C D AB	16 ways.
A	C	$\overline{A}$ B C D AB	16 ways.
A	—	$\overline{A}$ B C D AB	16 ways.

Question

Question Paper containing 10 True/False Questions, number of ways to attempt question paper such that:

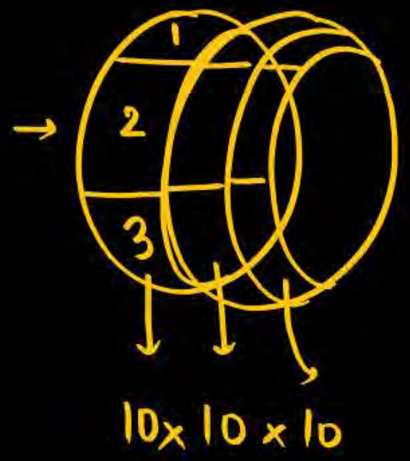
(i) No Restriction $\Rightarrow 3^{10}$.

(ii) All Questions should be answered. $\Rightarrow 2^{10}$.



Question

A letter lock consists of 3 rings each marked with 10 different letters. In how many ways, its is possible to make an unsuccessful attempt to open the lock?

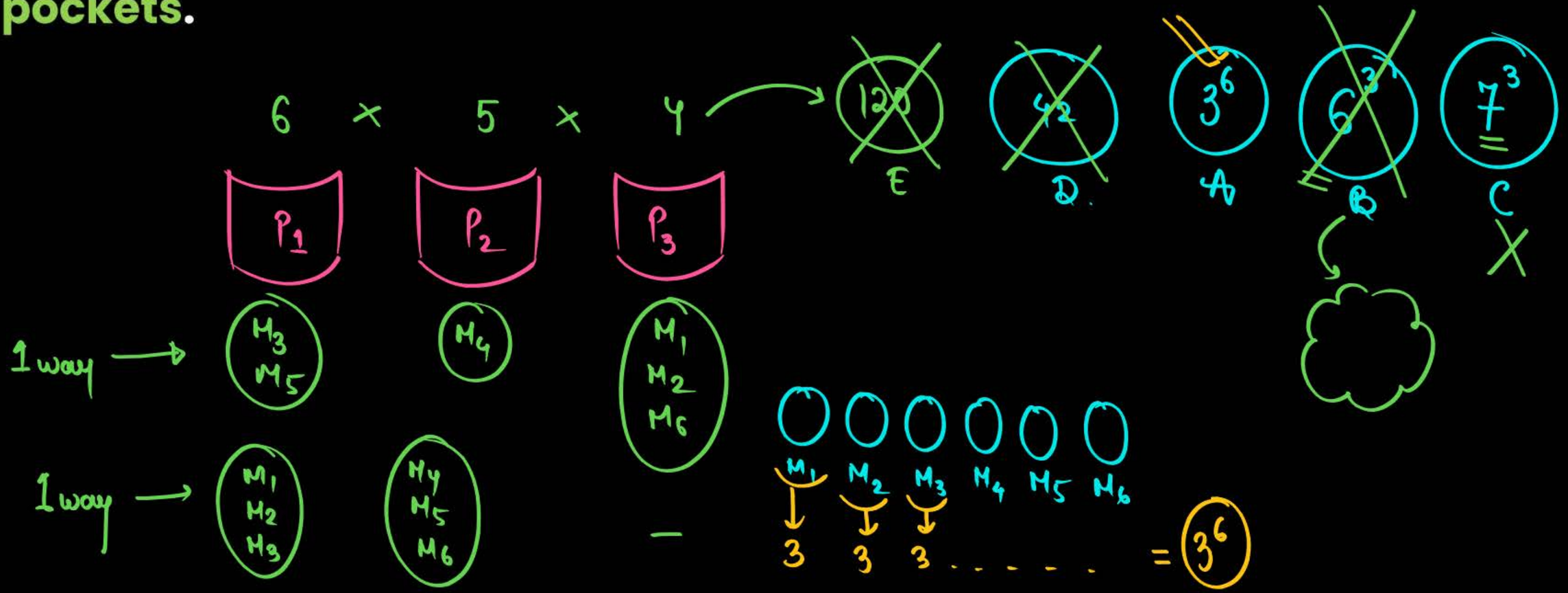


Total = 10^3.

- 1 correct
- 999 incorrect ✓

Question

A boy has 6 marbles, number of ways in which he can put these marbles in 3 pockets.



Question

Number of ways of posting 4 letters in 3 letters boxes such that:

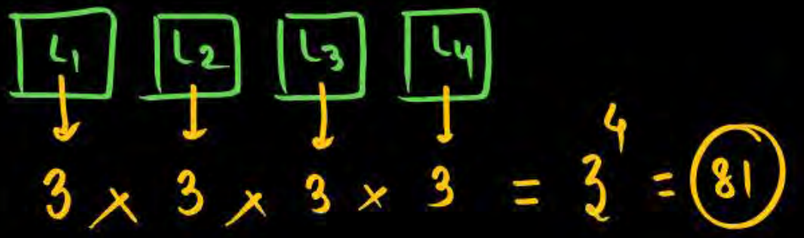
(i) No box has more than one letter

(ii) No box should contain all letters



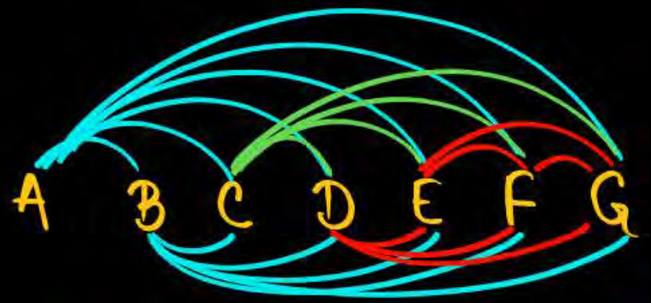
(i) zero.

(ii) $81 - 3 = 78$ ways.



Question

Total 7 person gathered in conference hall for Annual meet, find total number of Hand-shakes.



$6 + 5 + 4 + 3 + 2 + 1 = 21$

Question

Prove that (i) $\sin\theta + \sin(120° + \theta) + \sin(240° + \theta) = 0$

(ii) $\cos\theta + \cos(120° + \theta) + \cos(240° + \theta) = 0$

DIBY - (X)

Question

Prove that $\cos^2 x + \cos^2\left(x + \frac{\pi}{3}\right) + \cos^2\left(x - \frac{\pi}{3}\right) = \frac{3}{2}$

QiBY - Y



Homework

Re-attempt all Class Examples

DPP & DIBY

It is never about end result

It's always about Journey

Thank You

#futureIITians